

# Use Case CUD Product

## 1. Use case code

UC234

## 2. Brief Description

This use case describes the way how Product Manager interacts with AIMS System to add new, view, change information or delete media products like Books, Newspapers, CDs, DVDs,.. and manually adjust inventory quantities.

## 3. Actors

### 3.1 Product Manager

## 4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

## 5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select Add Product
  - 3.1. Software displays general form for Product Manager
  - 3.2. Product Manager input general information and choose specific type of product.
  - 3.3. Software displays form for fill specific information depending on product type
  - 3.4. Product Manager input specific information and submit.
  - 3.5. Software checks the data and price rule.
4. Product Manager select Edit Product
  - 4.1. Product Manager select one product and choose Edit
  - 4.2. Software displays product's current information
  - 4.3. The Product Manager update the necessary information
  - 4.4. Software check constraint of data and price

5. Product Manger select Delete Product
  - 5.1. Product Manager select up to ten products to delete at a time.
  - 5.2. Product Manager confirm the delete order
  - 5.3. Software check security limit delete.
  - 5.4. Software check product quantity
  - 5.5. Software remove from sale list
6. Product Manager select adjust quantity
  - 6.1. Product Manager select product to adjust
  - 6.2. Software display box of current stock and form for adjust
  - 6.3. Product Manager input new stock and reason
7. Software display notifications and save record into history

## 6. Alternative flows

| No | Location            | Condition                                                                        | Action                                                                               | Resume location    |
|----|---------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------|
| 1  | At step 3.5 and 4.4 | If the input data is not in the correct format                                   | Software display detailed error for each data field                                  | At step 3.4 or 4.3 |
| 2  | At step 5.1         | If Product Manager selects more than 10 products to delete at once.              | Software displaying the warning only allows a maximum of 10 products to be selected. | At step 5.1        |
| 3  | At step 5.3         | If total number of products that Product Manager deleted today over 20 products. | Software notify the Product Manager of a security violation and refused to delete    | At step 2          |
| 4  | At step 5.4         | If quantity of product removed greater than 0                                    | Software automatic change status to deactive.                                        | At step 7          |
| 5  | At step 6.3         | If Product Manager do not input Reason field                                     | Software require to enter reason for adjust product quantity                         | At step 6.3        |
| 6  | Any step            | If Product Manager cancel process                                                | Software do not save any changes and return product list interface                   | At step 2          |

## 7. Input data

| No | Data fields         | Description                                   | Mandatory    | Valid condition                           | Example                   |
|----|---------------------|-----------------------------------------------|--------------|-------------------------------------------|---------------------------|
| 1  | Barcode             | Unique identifier for the product             | Yes          | Unique in the system; numeric only        | 893123456789              |
| 2  | Title               | The name of the media product                 | Yes          | Cannot be blank                           | DVD Hello World           |
| 3  | Category            | Product classification                        | Yes          | One of: Books, Newspapers, CDs, DVDs      | Books                     |
| 4  | General Description | Condition, primary color, or return condition | Yes          | Text description                          | New condition, blue cover |
| 5  | Original Value      | Base price excluding 10% VAT                  | Yes          | Positive integer                          | 100                       |
| 6  | Current Price       | Selling price excluding 10% VAT               | Yes          | Between 30% and 150% of Original Value    | 120                       |
| 7  | Dimensions          | Physical size (Height x Width x Length)       | Yes          | Positive values only                      | 20 x 15 x 3 (cm)          |
| 8  | Weight              | Physical weight of the product                | Yes          | Positive value (kg)                       | 0.5                       |
| 9  | Author              | Creator(s) of the book                        | Yes (Book)   | Required if Category is "Books"           | Ngo Tat To                |
| 10 | Cover Type          | Format of the book cover                      | Yes (Book)   | "Paperback" or "Hardcover"                | Paperback                 |
| 11 | Editor-in-chief     | Head of the newspaper                         | Yes (News)   | Required if Category is "Newspapers"      | Nguyen Van A              |
| 12 | Artists             | Performers or singers                         | Yes (CD)     | Required if Category is "CDs"             | Son Tung M-TP             |
| 13 | Track List          | List of songs with titles and lengths         | Yes (CD)     | Must include title and duration per track | Track 1 - 10:30           |
| 14 | Disc Type           | Physical format of the disc                   | Yes (DVD)    | "Blu-ray" or "HD-DVD"                     | Blu-ray                   |
| 15 | Director            | Director of the media                         | Yes (DVD)    | Required if Category is "DVDs"            | Christopher Nolan         |
| 16 | New Stock           | Updated manual inventory count                | Yes (Adjust) | Non-negative integer                      | 50                        |
| 17 | Reason              | Explicit record for manual stock change       | Yes (Adjust) | Required for all manual adjustments       | Damaged during transit    |

## 8. Output data

| No | Data fields       | Description                                                                         | Display format     | Example                                              |
|----|-------------------|-------------------------------------------------------------------------------------|--------------------|------------------------------------------------------|
| 1  | Operation Status  | Notification of the result for the performed action (add, edit, delete, or adjust). | Text notification  | Product added successfully                           |
| 2  | Product List      | A list of all media products currently available in the store's inventory.          | Data table         | (List of Books, CDs, ...)                            |
| 3  | Product Details   | Full detailed information of a specific product when viewed or being edited.        | Form view          | Title: Prison Break, Price: 120,000,...              |
| 4  | Product Status    | The commercial availability of the product (available or deactivated).              | Label/Tag          | Deactivated                                          |
| 5  | History Log       | A record list of all product additions, edits, and deletions.                       | Timeline/Log entry | 16/03/2026: Manager A updated stock of Barcode X     |
| 6  | Error Message     | Detailed notification when input data is invalid or violates system rules.          | Text (Red)         | Price must be between 30% and 150% of Original Value |
| 7  | Stock Quantity    | The actual number of items in stock after a manual update or a sale.                | Positive integer   | 45                                                   |
| 8  | Adjustment Reason | An explicit record explaining why a manual stock change was performed.              | Text description   | Damaged during transit                               |